Array 22 (2024) 100345

Contents lists available at ScienceDirect

Array

journal homepage: www.sciencedirect.com/journal/array

A comprehensive review of explainable AI for disease diagnosis

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ARTICLE INFO

Keywords:
Interpretability
XAI
Deep learning
Machine learning
Disease diagnosis

ABSTRACT

Nowadays, artificial intelligence (AI) has been utilized in several domains of the healthcare sector. Despite its effectiveness in healthcare settings, its massive adoption remains limited due to the transparency issue, which is considered a significant obstacle. To achieve the trust of end users, it is necessary to explain the AI models' output. Therefore, explainable AI (XAI) has become apparent as a potential solution by providing transparent explanations of the AI models' output. In this review paper, the primary aim is to review articles that are mainly related to machine learning (ML) or deep learning (DL) based human disease diagnoses, and the model's decision-making process is explained by XAI techniques. To do that, two journal databases (Scopus and the IEEE Xplore Digital Library) were thoroughly searched using a few predetermined relevant keywords. The PRISMA guidelines have been followed to determine the papers for the final analysis, where studies that did not meet the requirements were eliminated. Finally, 90 Q1 journal articles are selected for in-depth analysis, covering several XAI techniques. Then, the summarization of the several findings has been presented, and appropriate responses to the proposed research questions have been outlined. In addition, several challenges related to XAI in the case of human disease diagnosis and future research directions in this sector are presented.

1. Introduction

AI is one of the most promising fields of computer science (CS), and research in this field is vibrant, with numerous areas for exploration, such as employing various techniques for detection and prediction tasks. Nowadays, the impact of AI has been seen on people's lives in different ways and the healthcare sector has witnessed a remarkable transformation driven by the rapid advancement of AI technologies recently [1].

In particular, ML and DL techniques have demonstrated their capability in several healthcare applications ranging from disease diagnosis [2–4] and drug discovery [5] to patient management and personalized treatment plans. Though AI-driven intelligent systems have shown outstanding performance and outperformed humans in performing certain analytical tasks in the field of healthcare, their complexity, lack of transparency and explainability have raised significant concerns and sparked criticism continuously. Therefore, applying AI applications in scientific contexts is difficult due to challenges in explaining how they works [6].

To achieve transparency, accountability, and comprehensibility from these intelligent systems, it is required to explain the ML and DL techniques' output efficiently. The need for these essential qualities in AI-based systems led to the evolution of the XAI field rapidly. It has already gained significant attention over the past few years due to its capability to make AI systems more trustworthy, more compliant, more efficient, and more robust [7]. This review article focusing on utilizing XAI in human disease diagnosis aims to present a thorough and critical analysis of the current status of research.

This paper analyzes the existing related articles, discusses the concepts of XAI and its applications, presents the merits and demerits of the XAI approaches, and mentions challenges and future research suggestions for it in the healthcare sector. Besides, an appropriate response to the following proposed research questions (RQs) based on analyzing several XAI-based healthcare-related research works that employ ML and DL techniques has been presented. The proposed RQs are listed below.

RQ1: Why is XAI important in human disease diagnosis, and what are some problems that can only be addressed with XAI but not with standard AI?

RQ2. What is the most frequently used XAI technique in ML and DL research, particularly in the context of human disease diagnosis?

RQ3: What are the potential strengths and limitations of different well-known XAI techniques?

RQ4: What are the main barriers and challenges in the case of widespread adoption of XAI techniques in healthcare settings, and how can these be overcome?

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https://doi.org/10.1016/j.array.2024.100345
Received 2 February 2024; Received in revised form 17 March 2024; Accepted 14 April 2024
Available online 26 April 2024

This review paper is sorted as follows: Section 2 presents the review methodology including the search strategy, inclusion and exclusion criteria, study selection procedure, and the data acquisition strategy to extract the most relevant information from the selected studies. Section 3 discusses the XAI concept from different perspectives. Section 4 presents the findings of an analysis of the final selected papers. The detailed responses to the proposed RQs are presented in Section 5. Section 6 lists several challenges related to XAI-based human disease diagnosis, including future research directions in this field. Section 7 presents a meaningful discussion of the selected papers, factors associated with the selection of different XAI techniques, and mentions the limitations of this review paper. Finally, Section 8 draws the conclusion of the review paper by mentioning the potential future work of this review.

2. Review methodology

To find the related studies, it is essential to follow a robust search strategy with meaningful query strings. After that, selecting the most appropriate studies among the related studies is also a paramount task to find out the most significant information from the studies. In this section, the adopted search strategy, inclusion and exclusion criteria, studies selection process, and data extraction process have been described.

2.1. Search strategy

The search is initiated with a total of 868 results by employing search strings, as tabulated in Table 1. The journal databases considered for this search are Scopus and the IEEE Xplore Digital Library. This review study is encompassed by all studies available up until September 27, 2023, from 2018.

2.2. Inclusion and exclusion criteria

The inclusion criteria to select the appropriate articles are listed below.

a. Studies that work with human disease classification/identification/recognition/detection.
b. Studies that consider at least one ML and/or DL model.
c. Studies that consider at least one XAI technique to interpret the result of the employed ML and/or DL models.

The exclusion criteria for selecting the appropriate articles are listed below.

a. The research works published before 2018 and after September 27, 2023.
b. Conference papers, books, magazines, and papers written in languages other than English.
c. Perspective, pre-editorial, review, and survey papers, along with inaccessible full-text studies and irrelevant studies.

Table 1
Query strings utilized to search the studies from two databases.

Database Name	Query String
Scopus	(ABS (explainability) OR ABS (explainable)) AND (ABS (machine AND learning) OR ABS (deep AND learning)) AND (ABS (medical) OR ABS (health) OR ABS (healthcare)) AND (LIMIT-TO (DOCTYPE, "ar"))
IEEE Xplore	(("Abstract":"explainability" OR "Abstract":"explainable") AND ("Abstract":"machine learning" OR "Abstract":"deep learning") AND ("Abstract":"medical" OR "Abstract":"health" OR "Abstract":"healthcare"))

d. Studies from journals categorized other than Q1 (based on the year 2022).

2.3. Study selection procedure

The PRISMA guideline was considered and followed to filter the search results systematically. The filtering procedure of the articles is presented in Fig. 1. First, 52 duplicate publications across two databases (Scopus and the IEEE Xplore Digital Library) were identified and excluded. Afterward, a screening process is conducted by reviewing the articles' titles and abstracts. At this stage, conference articles, books, magazines, and papers not written in English were excluded.

In the subsequent eligibility phase, perspective, pre-editorial, review, and survey papers, inaccessible full-text studies, and irrelevant studies were excluded. In addition, the quartile (Q) of the journals, which can be checked from the renowned scientific journal ranking websites (https://www.scimagojr.com/journalrank.php) was also considered for excluding the studies. Studies of journals that are categorized (based on the year 2022) as Q2, Q3, Q4, and journals not in any quartile were excluded. Consequently, 90 scientific papers were included for qualitative synthesis from the Q1 journal, which is renowned for its highest level of credibility.

For this paper, irrelevant studies are the following.

- Those works that are not primarily or not only related to human disease diagnosis
- Conceptual and theoretical studies without implementation
- Studies that do not consider either ML or DL techniques
- Studies that do not consider at least a single XAI technique
- Studies that incorporate additional technologies, such as blockchain and IoT, alongside ML and DL techniques
- Articles with poor/problematic study design
- Studies with unclear results and explanations
- Studies that do not consider evaluation metrics to assess the models

2.4. Data acquisition strategy

After completing the study selection procedure, extracting data and information from the finally selected articles is the next imperative task for the precise analysis and understanding of the study. To derive meaningful conclusions, some predefined attributes were considered, and many information from the selected articles was extracted and presented in tabular form. The first attribute, "Authors and Reference" refers to the authors' information and their corresponding reference number. The second feature of the table is the publication year of the articles. The third property presents the types of data used in the articles. The fourth attribute specifies the XAI techniques adopted in the papers. The fifth attribute reports the application or main task of the studies. Finally, the last attribute presents the cons of the paper/area of improvement of the finally selected paper. The findings of these preset attributes are presented in Section 4.

3. Explainable AI (XAI) techniques

XAI specifies the approaches utilized to construct AI systems that are transparent and interpretable by end users (i.e., domain experts, data scientists, or even people without academic knowledge about AI) [7]. XAI is not a monolithic concept; rather, it encompasses various interconnected ideas and notions. Explainability and interpretability are closely connected concepts: when systems are interpretable, they are considered explainable if their inner workings can be understood by persons. While 'explainable' is a prominent term in the field of XAI, the machine learning community tends to prefer using the term "interpretable" more frequently [8]. It is hard to define 'interpretability' mathematically. According to Tim Miller [9], interpretability can be defined as non-mathematically which is the level at which a person can

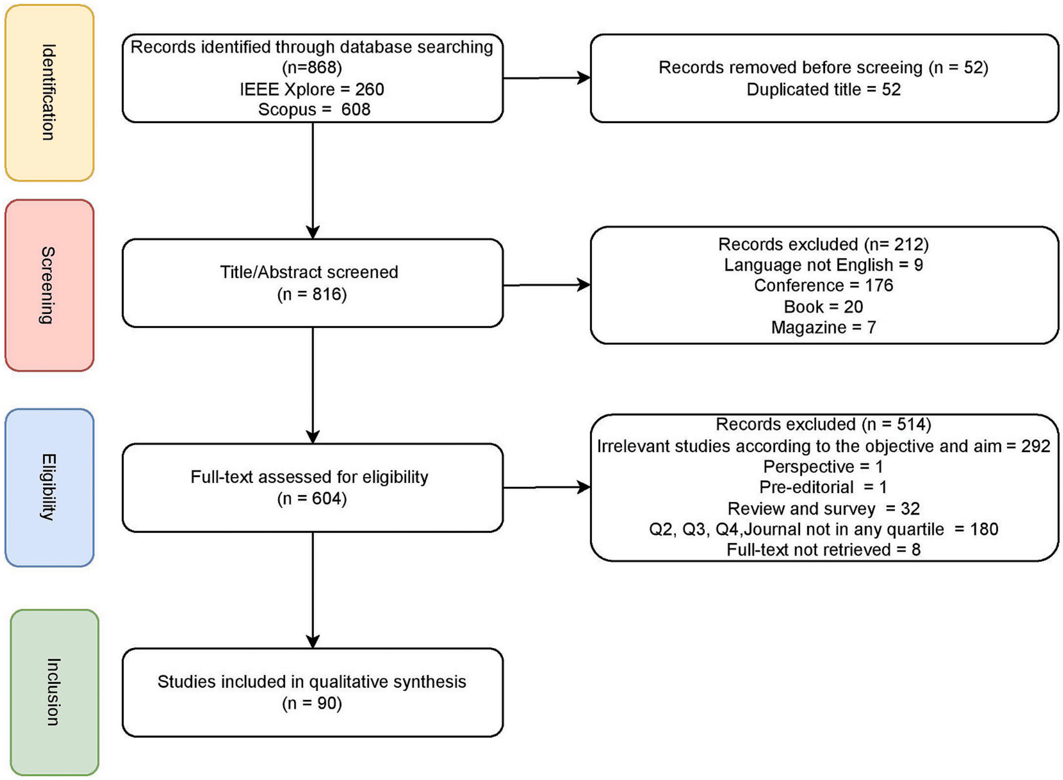

Fig. 1. Paper selection process using PRISMA guidelines.

comprehend the rationales for a particular output. According to Kim Been [10], interpretability is defined by the level at which a person can continuously predict the output of the models. In this section, the concept of XAI and its related terms have been discussed from different perspectives.

3.1. Need for XAI

XAI is essential and advancing, notably in the domain of the healthcare sector. It responds to the question "why" which results in increasing trust in AI. The need for XAI is presented in Fig. 2.

Explain to Justify: To justify the results of important real-world scenarios, XAI is utilized to provide information by maintaining the "Right to Explanation" standards that are specified in the General Data Protection Regulation (GDPR) [11].

Explain to Control: Justifying the models' decisions is not the only goal of explainability. This is also important to restrict the wrong interpretation and find the weaknesses and issues of the system.

Explain to Improve: Further rationale to construct an interpretable method is to develop it consistently by identifying the shortcomings. It is easily possible to improve an explainable and understandable model because users understand its functioning quickly and can identify ways to enhance its capabilities. So, XAI might facilitate continuous progress and promote collaboration between humans and machines.

Explain to Discover: Seeking explanations is an impactful approach to learning new facts, analyzing information, and increasing knowledge. Only systems that can be explainable are useful in this scenario.

In summary, explainability is a notably dynamic tool to explain AI-based decision-making processes. It helps in verifying models' predictions, improving models, and acquiring new perspectives. As a result, AI becomes more trustworthy [8]. In addition, explanations are classified into three elements, namely, source, scope, and depth [12].

3.2. Principles of explainable AI

A workshop with the larger AI community was organized where four XAI principles were proposed through the public comment period [13] which are presented in Fig. 3.

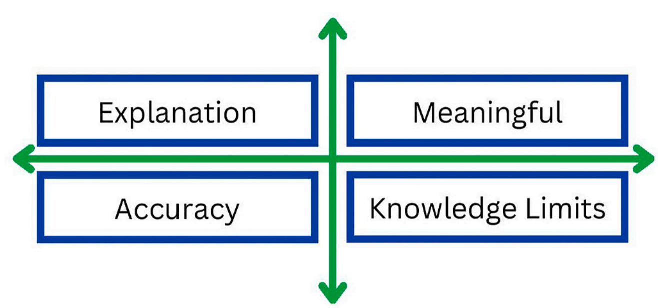

Fig. 3. XAI principles.

Fig. 2. Need for XAI [8].

Explanation: For particular outputs and/or processes, there is evidence or reason(s) provided by the system.

Meaningful: For the targeted clients, an understandable explanation is presented by the system.

Explanation Accuracy: An explanation appropriately represents why the system produces the particular output and/or adopts a particular procedure.

Knowledge Limits: The system functions only within its specified parameters and is operational only when it obtains enough degree of confidence in its results.

3.3. Properties of explanation

Explanation is characterized by two the following two properties.

Purpose: It describes the reason why someone asks for an explanation.

Style: It describes how an explanation is provided. It has three elements which are shown in the following Fig. 4. They are described below [13].

Level of detail: The level of detail ranges from sparse to extensive. In sparse, brief and minimal information is provided, and on the other hand, extensive provides a large amount of information.

Degree of interaction between human and machine: This is further partitioned into 03 categories: declarative explanations, one-way interaction, and two-way interaction. In a declarative explanation, the system offers only information without any interaction. One-way interaction is a higher degree of interaction. Here, the system gives an answer based on a received question. Lastly, the two-way interaction has the deepest interaction level that models a conversation between humans.

Format: It incorporates visual and graphical, verbal, and auditory or visual alerts.

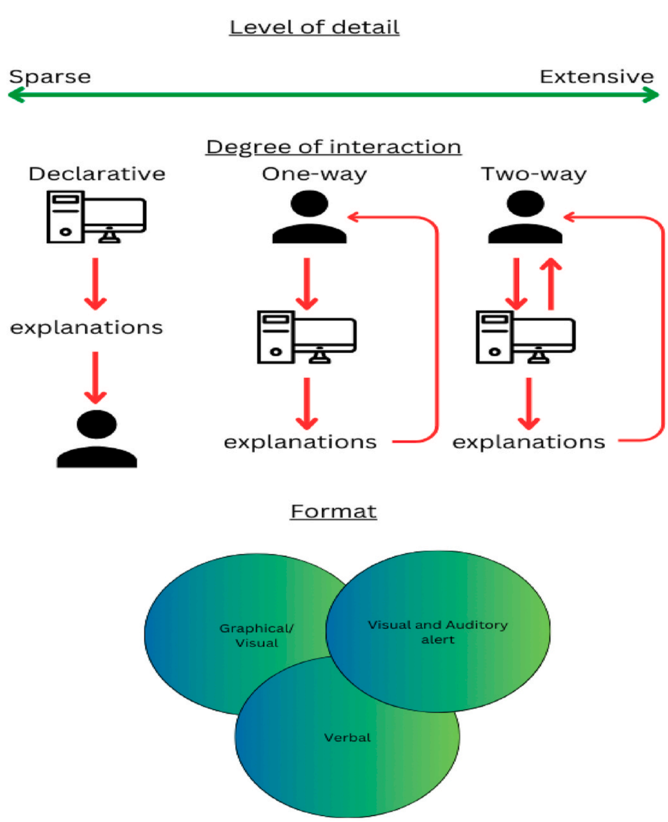

Fig. 4. Diagram of the elements of explanation styles.

3.4. Explainable AI and roles of stakeholders

There are different stakeholders (such as data scientists and ML developers, domain experts, and managers and business owners) engage with XAI. These three stakeholders play a significant role. XAI models are mainly responsible for producing the explanation for the ML models' inner workings and prediction. However, the roles of different stakeholders in XAI for a particular intelligent system are completely different from each other. Data scientists and ML model developers might use the explanation generated by the XAI techniques to get a better overview of the models' internal mechanisms and use this knowledge to optimize the models' performance. Field experts may incorporate whether the developed models' behavior is aligned with real-world scenarios or not. Based on the models' output, business owners and managers may consider the explanation to make informed decisions [14].

3.5. Tradeoff between interpretability and performance

The following Fig. 5 presents the models' accuracy vs its level of interpretability. From the figure, it is clear that models with higher accuracy require more interpretability, while models with lower accuracy are easy to understand but not very useful in many scenarios [15,16].

In general, transformer and deep learning models usually exhibit highly complex structures, while traditional ML models such as decision trees are not that complex to understand their internal structure and operations. Though the complex models achieve remarkable performance, their level of opacity is high (i.e., interpretability is low). On the other hand, less complicated structured models may perform slightly low but their level of opacity is low (i.e., interpretability is high). It is of note that based on the type of the working problems, models with less complex architecture(such as decision tree) may still show better performance than DL-based techniques while upholding the interpretability.

3.6. Classification of explainable AI techniques

Interpretability techniques can be classified into different categories. Each type of technique has its advantages and disadvantages. The following Fig. 6 represents the different types of interpretabilities.

Based on the scope, interpretability is split into two directions namely global explanation (explaining the overall model) and local explanation (explaining individual prediction). Based on the stage, it is split into ante-hoc (White-box models that are easily interpretable) and post-hoc (capable of explaining the previously trained model or its prediction). Then, post-hoc can be further classified into model-specific (the techniques that apply only to specific model types) and model-agnostic (the techniques that apply to all types of models). Some widely utilized example-based interpretability approaches are Counterfactuals, Prototypes and Criticisms, Adversarial, and Influential Instances. It focuses on demonstrating how the model behaves for individual cases [17].

4. Analysis and findings

In this section, finally selected 90 Q1 journals have been analyzed, and presented the findings from different perspectives into different subsections.

4.1. Evolution of Research: year-wise paper distribution

The number of articles concerning the year of publication is presented in the following Fig. 7. It is clearly shown that the number of scientific articles published has continuously risen in recent years, which indicates an increasing interest in this sector. The highest forty-five articles from the selected 90 Q1 journal studies were published in 2023. The trend to increase the number of studies each year is almost

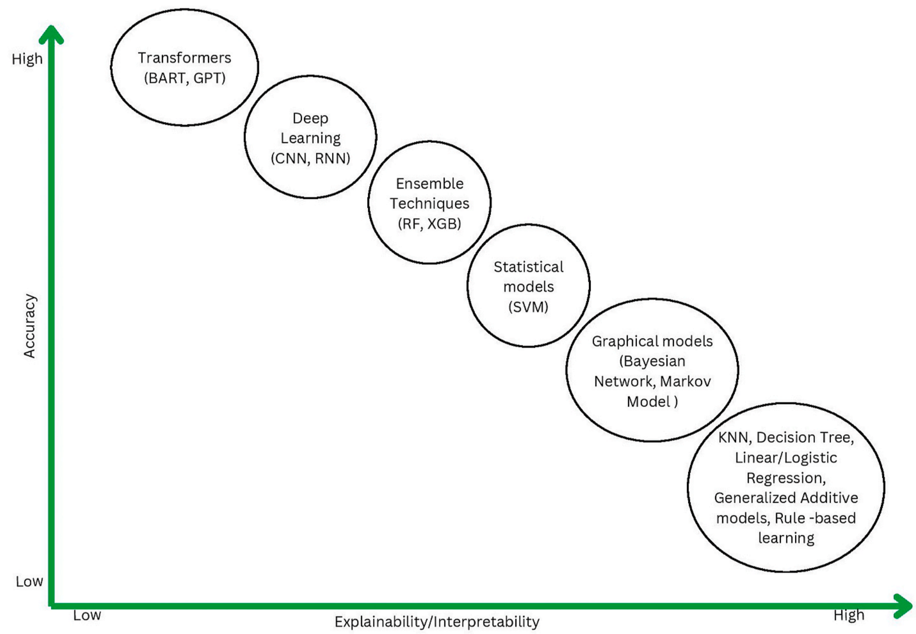

Fig. 5. Models' accuracy vs its level of interpretability.

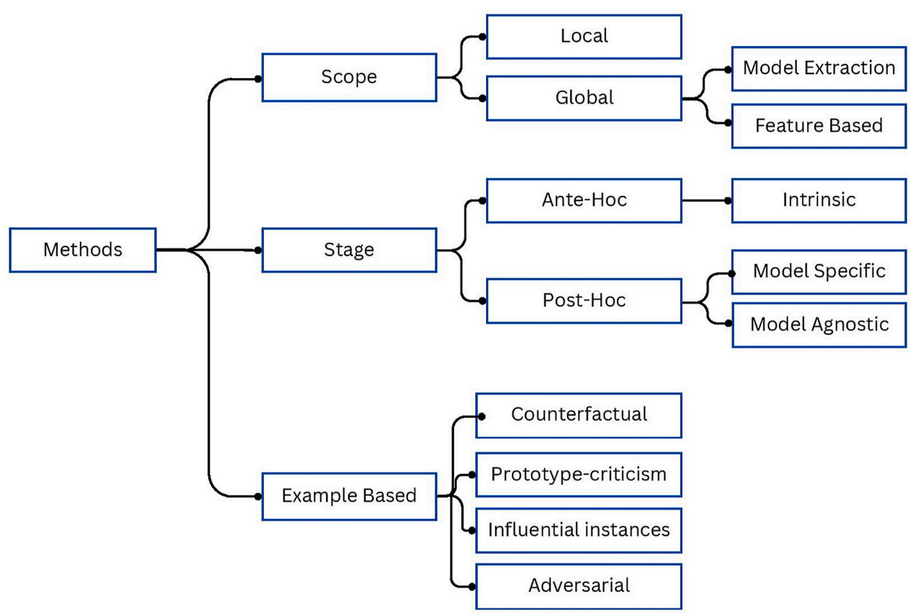

Fig. 6. Different types of interpretabilities [17].

twice.

Since the healthcare domain is very sensitive to work with, more research work should be conducted in this domain because it is essential to develop widely accepted XAI techniques to gain the highest level of trust among end users.

4.2. Keyword insights: word cloud analysis

To visualize the key concepts of the selected 90 papers, first, keywords or index terms from the papers are extracted, and after that, a word cloud is created and presented in Fig. 8. To create a word cloud, two libraries in Python such as Matplotlib and WordCloud have been utilized. In the following word cloud, each word size corresponds to its frequency in the papers. The most highlighted keywords or index terms are "deep learning", "machine learning", "explainable AI", and "artificial intelligence" which present the core topics of the research papers. In addition, the diverse range of keywords or index terms represents the multidimensional nature of ongoing research which provides a thorough overview of the present trends in the field of XAI-based human disease diagnosis.

4.3. Interpretable machine learning in human disease diagnosis

In this subsection, the article that employs XAI and ML techniques to diagnose human disease is considered only. Table 2 presents a summary of 29 articles that are related to disease diagnosis where ML and XAI techniques were used. It is observed that most of the papers in this category were published in 2023 and on the other side, the lowest number of articles in this category were published in 2021.

The most used data types are in tabular form (numeric data) here.

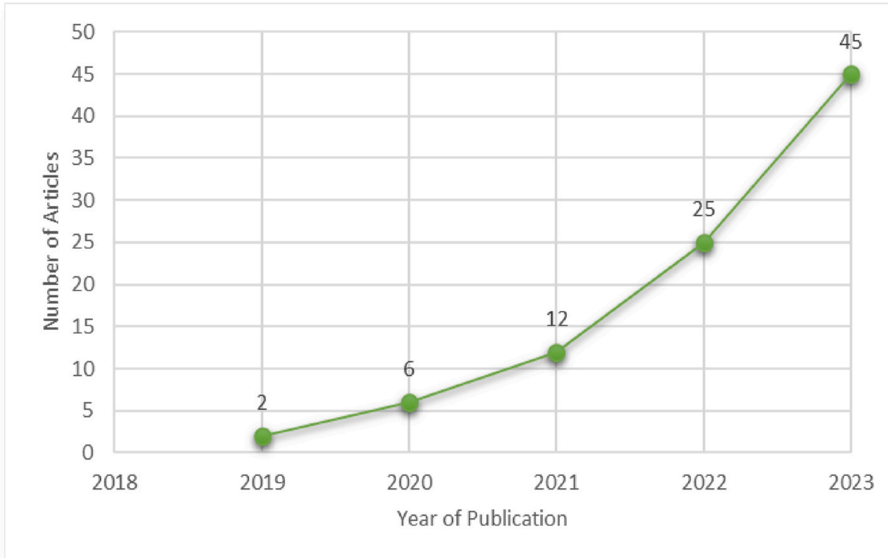

Fig. 7. The number of considered published articles over the years.

Fig. 8. Word cloud of paper keywords or index terms.

The XAI technique SHAP is mostly used to explain the ML models' output. The various ML and XAI techniques are used to identify, classify, and detection of various major human diseases including cancer, Parkinson's disease, hepatitis C, heart disease, depression, COVID-19, diabetes, dementia, stroke, kidney disease, Alzheimer's disease, and many others. The researchers of these research works are from different countries around the world.

4.4. Unveiling deep learning: explainability in human disease diagnosis

Explainability of DL techniques is a more difficult task than explainability of the traditional ML models due to the complex nature of the DL model's architecture and automated feature learning process. Table 3 presents a summary of 54 articles related to disease diagnosis where DL and XAI techniques were used. From the table, most of the research in this category (Where DL and XAI are utilized) among finally selected 90 research papers were performed in 2023, and the lowest number of the articles was published in 2019.

Most of the research works used various types of image datasets (MRI scans, X-ray images, ultrasound images, capsule endoscopy image, spiral and wave images, CT images, and many other types of images). In addition, some research works have utilized textual data. Most of the researchers adopted DL techniques utilizing class activation mapping techniques more specifically Grad-CAM to create a heatmap that highlights the input images' notable areas. The highlighted notable area is assumed to be the most significant in the particular class prediction, resulting in explaining DL models' output [51,53]. The various DL architecture and XAI techniques are used to predict, identify, classify, and detection of the various major human diseases including Alzheimer's disease, COVID-19, tumors, depression, glaucoma , various types of cancer, Parkinson's disease, skin diseases, several lung pulmonary diseases, tongue diseases, and many others.

4.5. XAI techniques in human disease diagnosis: application with machine learning and deep learning

Among the finally selected papers, there are few articles where both ML, DL, and XAI techniques are utilized and those are presented in this

Table 2
Summary of various literature related to human disease diagnosis where ML and XAI techniques were used.

Authors and Reference	Article Publication Year	Data Types Used	XAI Techniques	Application/Main Task	Cons of the Work/Area of Improvement
Islam et al. [18]	2023	Tabular data	SHapley Additive exPlanations (SHAP)	Predicting the risk of hypertension	Relatively small dataset size, consisting of only 612 instances
Silva-Aravena et al. [19]	2023	Tabular data	SHAP	Breast cancer and the risk factors identification	Relatively small dataset size, consisting of only 400 instances
Zhu et al. [20]	2023	Tabular data from Electronic medical records (EMR)	SHAP and Breakdown plot	Differentiate bipolar disorder from major depressive disorder	i. Class imbalance problem ii. Genetic risk factors and neuroimaging-based biomarkers were not taken into consideration
Huang et al. [21]	2023	Tabular data	SHAP	Ovarian cancer diagnosis	The score of AUC-ROC attained by the most optimized random forest (RF) techniques is not high enough and is only 71.00 %
Deshmukh et al. [22]	2023	Tabular data	Local Interpretable Model-Agnostic Explanations (LIME)	Medical data classification	Relatively small dataset size, consisting of only 600 records for breast cancer dataset and 510 records for the knee MRI dataset
Ding et al. [23]	2023	Tabular data	SHAP	Cardiac health risk determination	i. Inaccurate assessment of residential greenness exposure at the individual level ii. The cross-sectional design limits the establishment of causal associations between cardiac abnormalities and residential greenness
Junaid et al. [24]	2023	Multimodal time-series data	SHAP, LIME, and Shapley Additive exPlanations for Automated Summary Help (Shapash)	Parkinson's disease detection at an early stage	Loss of information during the process of turning the time-series modality into a non-time series by extracting statistical features across patients' visits
Fan et al. [25]	2023	Multivariate tabular data	SHAP, LIME with Stability	Prediction of hepatitis C disease	i. Relatively small dataset size ii. More factors such as lifestyle habits should be considered
Chang et al. [26]	2023	Tabular data from electronic health records (EHR)	Shapley Additive exPlanation (SHAP) values	Predict pediatric Acute respiratory infection pathogens at admission	i. Possibility of compromising the model's robustness ii. It is possible to show some selection bias in the experimental results
Bernard et al. [27]	2023	Tabular data	Partial Dependence Plots (PDP) and SHAP	Predicting personalized physiological aging	It is possible that findings may not generalize to populations outside the scope of the National Health and Nutrition Examination Survey (NHANES).
Kasani et al. [28]	2023	Tabular data	ELI5, PDP, LIME and SHAP	Classification of clinical depression and evaluation of nutritional status	The work was a single-race and retrospective study
D'Angelo et al. [29]	2022	Tabular data	LIME and SHAP	Patterns identification in multiple biomarkers for diabetic foot diagnosis	Relatively small dataset size
Moreno-Sánchez et al. [30]	2023	Tabular data	Permutation Feature Importance, PDP, and SHAP	Heart failure survival prediction	Possibility of the bias in model's performance
Daluwatte et al. [31]	2023	Clinician notes and electronic healthcare records	SHAP	Finding pertussis episodes	Lack of generalizability of the proposed method
Sharma et al. [32]	2023	Wavelet-Based Single-Channel electroencephalogram (EEG) Signal	SHAP	Cyclic Alternating Pattern (CAP) Phases and Sub-Phases Detection	This work needs to be tested using a diverse and large dataset
Basta et al. [33]	2022	Tabular data	PDP and SHAP	Mild Cognitive Impairment diagnosis	The cross-sectional nature of the dataset used for model training and cross-validation
Kırbŏga et al. [34]	2023	Tabular data	SHAP	Diagnosis of Troponin levels in COVID-19 patients	Performance can affected by sparse and unstructured data
Yadav et al. [35]	2023	Tabular data	LIME and SHAP	Prediction of Non-Communicable Chronic Disease	SMOTE has been used before dataset splitting (training and testing data) resulting in data leakage problems
Mridha et al. [36]	2023	Tabular data	LIME and SHAP	Development of automated Stroke prediction	No feature selection was performed
Moreno-Sánchez et al. [37]	2023	Tabular data	Permutation Feature Importance, PDP, and SHAP	Early Diagnosis of Chronic Kidney Disease	There is a lack of external validation to substantiate objective experimentation
Li et al. [38]	2022	Tabular data	SHAP	Identification of coronary heart disease	Possibility of bias in results
Chalabianloo et al. [39]	2022	Physiological sensor data	SHAP	Classification of stress	Inability to gather real-life data
Bogdanovic et al. [40]	2022	Tabular data	SHAP	Alzheimer's disease diagnosis	To balance the dataset, sampling techniques have been used on test data
Vyas et al. [41]	2022	Clinical data	LIME	Identifying the presence and severity of dementia	A comprehensive assessment has not been performed across different populations

(*continued on next page*)

Table 2 (*continued*)

Authors and Reference	Article Publication Year	Data Types Used	XAI Techniques	Application/Main Task	Cons of the Work/Area of Improvement
Weng et al. [42]	2022	Tabular data	SHAP	Distinguish between Crohn's disease and intestinal tuberculosis	Relatively small dataset size and collected from a single center
Kibria et al. [43]	2022	Tabular data	SHAP, LIME, and Permutation Feature Importance	Prediction of Diabetes Mellitus	The model lacks capability in a real-life medical setting
Islam et al. [44]	2022	EEG recordings	ELI5 and LIME	Stroke prediction	This work does not consider multimodal biosignal data
El-Sappagh et al. [45]	2021	Tabular data	SHAP	Alzheimer's disease diagnosis	The approach is not applicable in a real-world clinical setting
Davagdorj et al. [46]	2021	Tabular data	DeepSHAP (Proposed)	Non-Communicable Diseases Prediction	This work has been performed and analyzed at only a specific point in time

subsection. Table 4 presents a summary of various papers related to disease diagnosis where XAI techniques were used along with ML and DL models.

Several types of data were used by the researcher in this case such as text data, ECG data, and image data. More than half of the paper was published in this case in 2023. The LIME, Integrated gradients, Grad-CAM, Attention mechanism, and shapely values are the different types of XAI approaches utilized in these papers. Significant application areas are COVID-19 prediction, detection of depression levels, and assessment of psychological disorders.

5. Response to the proposed research question (RQs)

An appropriate response to the four proposed RQs is presented in this section. Preparing the utmost response to the proposed RQs by analyzing the existing related works was one of the targets because it will help to understand the current scenario of this field. After extensive analysis of the selected and relevant papers, a detailed response to the proposed RQs is prepared below.

RQ1: Why is XAI important in human disease diagnosis, and what are some problems that can only be addressed with XAI but not with standard AI?

The role of the XAI is significant in gaining the trust of end-users. Human disease diagnosis is a very sensitive issue and employing the predictive models in this regard without full confidence or trust is not feasible. Black box AI or standard AI is unable to provide the rationale behind the models' prediction or decision. Sometimes standard AI is biased while training the models because of the lack of a diverse dataset which results in trust issues. Therefore, standard AI causes unjustifiable and unaccountable output [17]. In this point, Explainable AI provides an explanation and interface to tell the reasons behind the models' decisions. Consequently, predictive models achieve the trust of the user. The following Fig. 9 presents the difference between blackbox AI and XAI.

RQ2: What is the most frequently used XAI technique in ML and DL research, particularly in the context of human disease diagnosis?

From the analysis of Section 4, it is seen that overall, SHAP and Grad-CAM are the two most widely utilized XAI techniques. SHAP is applied to explain which clinical symptoms are significant for the accurate diagnosis of particular diseases. In addition, Grad-CAM is the most widely used to provide visual explanation via heatmap by highlighting and emphasizing the notable areas of input images in medical settings. In addition, LIME is also considered in many research works to explain the models' output.

RQ3: What are the potential strengths and limitations of different well-known XAI techniques?

Each of the existing XAI techniques has several strengths and limitations, some of which are presented in Table 5. Selecting XAI techniques depends on the particular requirements of the working problem, considering factors such as the models' complexity, the dataset characteristics, and the preferred level of explainability. Moreover, combining several techniques or utilizing hybrid approaches can be more beneficial and provide more insights into models' behavior.

RQ4: What are the main barriers and challenges in the case of widespread adoption of XAI techniques in healthcare settings, and how can these be overcome?

Trusting the results produced by powerful predictive models without explanation is no longer acceptable today because of several issues such as biased datasets, reliability, credibility, and adversarial instances in artificial intelligence [14]. The widespread adoption of XAI techniques in healthcare has been delayed due to the lack of an in-depth evaluation of the explanations of effectiveness created by XAI techniques [110]. Another challenge and barrier is the lack of standardized methods to explain the AI model's output. To overcome the barriers and challenges in the widespread adoption of XAI in healthcare, developing more interpretable models, and user-friendly and cost-effective standardized XAI tools, specific guidelines for the use of XAI in healthcare settings are required.

6. Challenges and future research directions

To develop robust XAI techniques, finding the challenges existing research works face is essential. Based on the challenges, different types of solutions can be proposed. In this section, firstly, different types of challenges have been discussed, and after that, possible areas where the current researcher should be focused have been listed.

6.1. Challenges

Current privacy law emphasizes the transparency of health-related data, and the AI models that incorporate these data should be also transparent and more explainable. Creating an AI model explainable is a challenging task. Day by day, the newly developed AI models are becoming more complex with increasing performance [15,16]. Consequently, it creates difficulty to understand by the end users (patients) and makes it challenging to keep trust. However, the primary aim should not likely be to explain these models to the end user but to the researcher and data scientist involved with developing the new models. End users such as patients mostly expect the prediction made by the AI models should be highly accurate. Another important challenge is to keep a balance between accuracy and interpretability. In the case of human disease diagnosis, accuracy (performance) is more vital than explainability as it might be a matter of death of human beings. So, interpretability might be thought of as less significant in disease diagnosis than accuracy (performance) [111].

There is a lack of trust and interpretability of DL models in the case of performing accurate diagnosis. Through efficient XAI techniques, it is possible to provide a higher level of transparency in DL models without sacrificing the privacy of data [100].

Though it is almost established that CNNs have performed

Table 3
Summary of various literature related to disease diagnosis where DL and XAI techniques were used.

Authors and Reference	Article Publication Year	Data Types Used	XAI Techniques	Application/Main Task	Cons of the Work/Area of Improvement
Shojaei et al. [47]	2023	MRI scans	Occlusion Map and Backpropagation-based methods	Alzheimer's disease diagnosis	Relatively small dataset size
Liz et al. [48]	2023	X-ray images	Gradient-weighted Class Activation Mapping (Grad-CAM)	Understanding and classification of multilabel imbalanced Chest X-ray datasets	Many other different strategies can be employed to alleviate the data imbalance problem
Loh et al. [49]	2023	Electrocardiogram (ECG) signal	Grad-CAM	Automated detection of Attention Deficit Hyperactivity problem and Conduct Disorder	Relatively small dataset size and class imbalance problem. Consequently, lack of generalizability of the proposed method
Resendiz et al. [50]	2023	Images	Grad-CAM, GRADCAM++, Hi-Res-CAM, Xgrad-CAM	Acute Lymphoblastic Leukemia classification	Exploration of unsupervised or deep reinforcement learning is needed
Ibrahim et al. [51]	2023	Imaging data of bone scintigraphy (BS)	Grad-CAM	Detection of metastatic bone disease	Prospective validation is necessary before the algorithm can be implemented in clinical settings
Ukwuomaa et al. [52]	2023	X-ray images	Grad-CAM to produce saliency maps and attention mechanism	COVID-19 prediction	Security guidance is required due to susceptibility to adversarial attacks
Mercaldo et al. [53]	2023	Magnetic resonance imaging (MRI)	Grad-CAM	Brain Cancer Detection and Localization	Lack of consideration for cancer-grade detection
Le et al. [54]	2023	Large-scale multi-class ECG dataset	Lead-wise Grad-CAM	3-lead Electrocardiogram classification	It is impossible to identify certain types of cardiovascular abnormalities
Shin et al. [55]	2023	Multiparametric MRI data	Layer-wise relevance propagation (LRP)	Discrimination of tumour	Clinical benefit needs to be more investigated
Ikechukwu and Murali [56]	2023	X-ray images	SHAP and Grad-CAM	Chronic obstructive pulmonary disease diagnosis	Lateral chest x-ray images are excluded.
Rahman et al. [57]	2023	2D ultrasound images	LIME and Grad-CAM	Identifying stages of fetal development	Still, there are more areas to investigate.
Lampert et al. [58]	2023	ECG signal	Grad-CAM	Cardiomyopathy prediction	Introducing selection bias
Chattopadhyay et al. [59]	2023	Histological images	Grad-CAM	Colorectal cancer prediction using histological images	Relatively low performance for certain classes
Mercaldo et al. [60]	2023	MRI data	Grad-CAM	Alzheimer Classification and Localization	Accuracy is low compared to existing works.
Saravanan et al. [61]	2023	Spiral and Wave images	LIME	Early Prediction of Parkinson's Disease	A limited amount of hand-drawing dataset
Yang et al. [62]	2023	Tabular data	SHAP	Invasive bacterial infection prediction in febrile infants	Relatively small and single-center dataset
Corbin et al. [63]	2023	Images	Grad-CAM	Skin lesion classification	Fairness in the model predictions can be improved
Dissanayake et al. [64]	2023	ECG signal	Neural conductance visualizations and SHAP	Identification of cardiovascular diseases	A clear difference between the train and test losses/accuracies
Mukhtorov et al. [65]	2023	Images	Grad–CAM	Endoscopic Image Classification	A lack of updated open-source datasets
Song et al. [66]	2023	Ultrasound images	Grad-CAM and GSInquire	Detection of COVID-19 cases	Data quality and the dataset size can be improved
Dong et al. [67]	2022	EHR data	Shapley value	Opioid overdose risk prediction	The outcomes might not generalize effectively to individuals who utilize non-prescribed opioids
Naz et al. [68]	2023	Images	LIME	Classification of several lung pulmonary diseases	The presence of inaccuracies in the equation
Altan [69]	2022	Optical coherence tomography (OCT) images	Feature activation maps	Analysis of macular edema on OCT images	The diverse medical devices used in OCT imaging
Kato et al. [70]	2022	Computed Tomography (CT) images	Grad-CAM and Attention map	Classification and visualization of pneumonia induced by COVID-19	Depends on an infection segmentation mask during the training phase
Sheu et al. [71]	2023	X-ray images	SHAP and Grad-CAM	Classification of Pneumonia Infection	Security Concerns
Prezioso et al. [72]	2022	CT images	Attention map	Segmentation and classification of salivary gland tumors	Variability in CT scan resolutions might need upsampling or downsampling, which causes a loss of valuable information
Bhandari et al. [73]	2022	X-ray images	SHAP, LIME, and Grad-CAM	Classification of COVID-19, Pneumonia and Tuberculosis	Relatively small dataset size; Therefore model's performance needs to be tested on the extensive dataset
Soliman et al. [74]	2022	Brain scans images	Occlusion maps	Classification of neurodegenerative disorders	Relatively small dataset size
Hamza et al. [75]	2022	X-ray images	Grad-CAM	Classification of COVID-19 cases	It takes longer to calculate the classification accuracy after the fusion process
Abbas et al. [76]	2022	Images	Grad-CAM	Tumour Image Classification	Data irregularities

(*continued on next page*)

Table 3 (*continued*)

Authors and Reference	Article Publication Year	Data Types Used	XAI Techniques	Application/Main Task	Cons of the Work/Area of Improvement
Sharma and Mishra [77]	2022	X-ray images	Grad-CAM	Diagnosis and severity assessment of COVID-19	Lack of generalizability and interpretability for the whole slide images
Zhang et al. [78]	2021	Images	Attention map	Tongue disease diagnosis	Relatively small dataset size
Jahmunah et al. [79]	2022	ECG signal	Grad-CAM	Detection of myocardial infarction	The dataset that has been utilized in this study is imbalanced and not authentic hospital data
Khan et al. [80]	2022	X-ray images	Grad-CAM	COVID-19 classification	The fixed threshold value used in the optimization algorithm
Li et al. [81]	2021	Images	SHAP	Brain Tumor and Mitosis Detection	Leveraging clustering technologies for feature extraction from images might not be suitable for dynamic scenarios
Abir et al. [82]	2022	Images	LIME	Diagnosis and Anticipation of Leukemia	Accuracy is low compared to existing works.
Liu et al. [83]	2021	CT images	Grad-CAM	Microvascular invasion prediction in hepatocellular carcinoma	The dataset size used for developing the model in this study is comparable but not significantly larger than previous studies
Singh and Sharma [84]	2022	ECG signal	SHAP, Grad-CAM, LIME, K-GradCam (Authors proposed)	Classification of Arrhythmia	The interpretation method fails to generate heat maps for a few instances
Shorfuzzaman et al. [85]	2021	Images	Grad-CAM and SHAP	Diagnosis of diabetic retinopathy grading	Due to the presence of various imaging conditions, the model performs misclassifications to a great extent
Thakoor et al. [86]	2020	Optical Coherence Tomography (OCT) images	Testing with Concept Activation Vectors (TCAV) and Grad-CAM	Glaucoma detection	May overlook valuable patterns or concepts
Hughes et al. [87]	2021	ECG signal	LIME	Predict the presence of thirty-eight diagnostic classes in five groups	External validation has not been performed
Kamal et al. [88]	2022	Images	Sub-modular LIME	Prediction of Glaucoma	Relatively small dataset size
Nafisah and Muhammad [89]	2022	X-ray images	Grad-CAM	Detection of Tuberculosis	The system's generalization ability can be improved
Hou et al. [90]	2021	Images	SHAP	Ischemia-reperfusion injury prediction	Lack of generalization ability of the model
Kumar et al. [91]	2021	MRI data	Class Activation Mapping (CAM)	Diagnosis of brain tumor	Secure data transmission should be considered
Uddin et al. [92]	2021	Text data	LIME	Prediction of depressive symptoms	The level of depression has not been considered in the work
Sousa et al. [93]	2019	Images	LIME	Classification of lymph node metastases	Medical experts' opinions should be considered
Chang et al. [94]	2020	Images	Adversarial Example and Grad-CAM	Focusing on glaucoma and glaucoma-related findings.	The dataset has little diversity compared to other related studies and a lack of generalizability of the model
Barata et al. [95]	2020	Images	Attention map and Activation map	Diagnosis of skin lesion	The robustness of the extracted features can be improved
Laiz et al. [96]	2020	Capsule endoscopy images	Class Activation Mapping (CAM)	WCE polyp detection	The dataset is highly imbalanced
Singh et al. [97]	2021	Images	Heatmap	COVID-19 screening	Image preprocessing techniques can be considered due to noise and visibility issues in the images
Hemelings et al. [98]	2020	Images	Saliency map	Prediction of glaucoma from colour fundus images	Class distribution, data Imbalance, and disease stage representation problems in the dataset
Lee et al. [99]	2019	CT images	Heatmap	Detection of acute intracranial haemorrhage	Lack of generalizability of the system and the possibility of bias
Varam et al. [100]	2023	Capsule endoscopy image	Grad-CAM, GradCAM++, LayersCAM, LIME, and SHAP	Classification of Wireless Capsule Endoscopy image	Verification of results through medical practitioners is necessary

outstanding, their application in the healthcare domain is still confined in many scenarios due to several reasons (poor and small medical database size since acquiring data requires high costs, complexity in labeling medical data, and confidentiality issues to access and utilization of these data) [96]. Therefore, lack of a standard dataset is another challenge in the field of disease diagnosis. Due to these reasons, models are not performing as desired and suffer from various problems such as underfitting [97].

Different ML and DL models have different levels of complexity and opacity. Some techniques have greater explainability than others. There is a tradeoff between interpretability and accuracy: the highly performed models are less interpretable (such as CNN and RNN) and the highly interpretable models show less performance (Such as linear or logistic regression) [15,16]. Instead of being fixed at this point, achieving high accuracy and high interpretability together is a dynamic goal that current researchers aim to achieve [8].

Some post-hoc explanation methods have their limitations for interpreting time-series problems (repetitive occurrences of the identical feature may result in unreliable explanations) namely SHAP and Grad-CAM. To solve this, a novel K-GradCam technique has been introduced that combines the advantages of gradient-based and perturbation-based techniques [84].

Table 4
Summary of various literature related to disease diagnosis where XAI techniques were used along with ML and DL models.

Authors and Reference	Article Publication Year	Data Types Used	XAI Techniques	Application/Main Task	Cons of the Work/Area of Improvement
Solayman et al. [101]	2023	Tabular data	LIME	Automatic COVID-19 prediction	Use of data from a particular region, and this work needs to be tested using diverse datasets
Siddiqua et al. [102]	2023	Tabular data collected through a questionnaire	LIME	Depression level detection and prediction among university students in Bangladesh	The dataset has an extensive feature set (106 questions), and the work only considers university-going students. So, it may not show reliability for the population with different profiles (for example, children and elderly)
Alavijeh et al. [103]	2023	Text data (Twitter)	Integrated gradients	Assessment of psychological disorders	Lack of reliability of data
Rajeshwari et al. [104]	2023	Phonocardiogram signal	Grad-CAM	Phonocardiogram events detection associated with mitral valve prolapse	In this work, the decision made by the 2D convolutional neural network (CNN) lacks full explainability and not convincing enough within the clinical community
Singh et al. [105]	2021	X-ray images	Grad-CAM	COVID-19 diagnosis	Relatively small dataset size
Zogan et al. [106]	2022	Text data	Attention mechanism	Multi-Aspect Depression Detection	Analyzing users' tweets for both topics and sentiments simultaneously has not been performed
Ibrahim et al. [107]	2020	ECG data	Shapley value	Prediction of Acute Myocardial Infarction	This work only used one ML model and two DL models

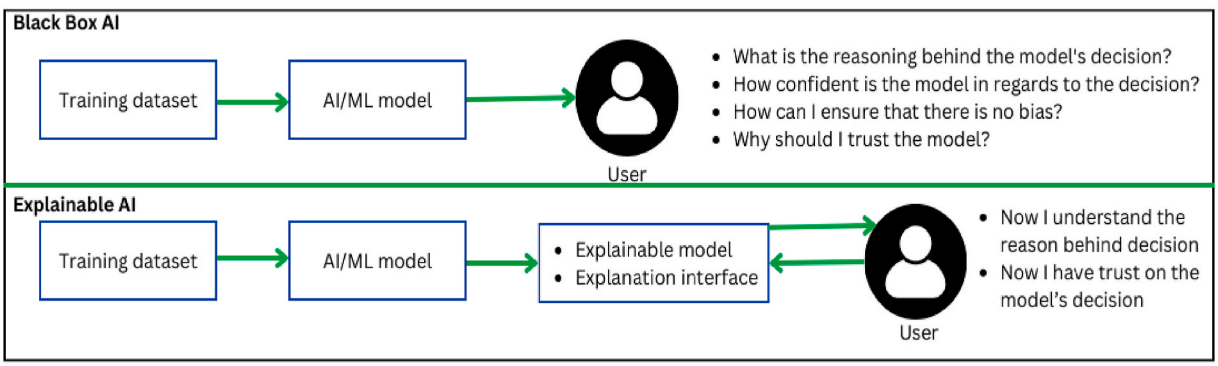

Fig. 9. Standard AI behaves as black-box models whereas XAI shows transparency.

Different stakeholders are involved with XAI and their requirements of interpretability are different [14]. As a result, developing different explanations for different stakeholders is a potential challenge for XAI.

Lastly, significant growth was observed in XAI in recent years, with some strong principles established, despite that the "explainability" issue has not been deemed fully addressed [57]. In addition, each of the XAI techniques has some potential limitations that are required to be addressed [15,17,108,109].

6.2. Future Research directions

The field of XAI is still evolving. The scientific community is exploring this field from various directions to increase the interpretability, transparency, and trustworthiness of AI systems. Some potential future research directions of XAI are presented below.

a. The researcher should only work on an extensive and diverse dataset that is clinically validated by domain experts to gain trust and accountability in their work.
b. Establishing collaboration among researchers from different disciplines such as CS, psychology, ethics, medicine, etc. needs to be emphasized for the validation of the research work.
c. Researchers should focus on the integration of ethical concerns with XAI techniques that help to ensure fairness, accountability, and transparency. They may also focus on exploring XAI techniques that meet with proper ethical guidelines.
d. Researchers should emphasize establishing standardized methods to evaluate the effectiveness of the XAI techniques. It will help to assess the model's interpretability more effectively and make it easier to compare and measure the advancement of this field.
e. Researchers should emphasize developing XAI models that are capable of providing different types of explanations for different stakeholders.
f. Researchers should perform extensive research and create protocols for assisting explanation-seeking dialogues between end users and artificial intelligence systems [112].

7. Discussions and limitations

Based on a deep investigation of the selected 90 articles, it is observed that there are many areas of improvement, such as the small dataset size and single-center dataset, scarcity of diverse datasets, lack of reliability of the data, information loss, containing data imbalance problems, lack of external validation, lack of generalizability of the models, security issues, lack of clinical benefits and lack of result verification through medical practitioners, algorithmic bias, selection bias, bias in model's performance, lack of fairness, lack of robustness of data and models, lack of explainability of DL models, and lack of model's performance in real-life medical settings. Since the application of AI in healthcare settings needs to be precise and accurate without a small margin of error, it is therefore required to solve the abovementioned issues precisely before adopting it in real-time clinical settings.

Choosing an appropriate XAI technique is challenging. The following Table 6 describes different factors/considerations associated with particular XAI techniques.

This review paper has three limitations. Firstly, due to the limited

Table 5
Strengths and limitations of several well-known XAI techniques [15,17,108,109].

Name of XAI Techniques	Advantages/Strengths	Disadvantages/Limitations
Partial Dependence Plots (PDP)	● The computation is intuitive. ● Interpretation is clear for uncorrelated cases. ● Easy to implement. ● The calculation has a causal interpretation.	● Some PDPs do not show the feature distribution. ● Independence assumption is the big issue. ● Heterogeneous effects might be hidden. ● More complicated for correlated features.
Accumulated Local Effects (ALE)	● Unbiased plots. ● Faster to compute than PDPs. ● The interpretation is clear. ● The complete prediction function can be broken down into a summation of smaller-scale ALE functions.	● Interpreting the effect across intervals becomes challenging and is not allowed in instances of strongly correlated features. ● Plots can become a bit shaky with a high number of intervals. ● Second-order effect plots can be challenging to understand to a certain extent. ● Compared to PDPs, it is less intuitive and the implementation is highly complex.
Permutation Feature Importance	● Offers an extremely compressed and global view of the model's actions. ● It considers all interactions with other features. ● No need for the model's retraining.	● Associated with the model's flaws. ● Access to the true outcome is essential to calculate the permutation feature importance. ● In the case of correlated features, it may show biases. ● The output might vary to a great extent in the case of repeated permutations.
Individual Conditional Expectation (ICE)	● More intuitive to understand than PDPs. ● Uncover heterogeneous relationships. ● Highly specific.	● Only display a single feature meaningfully. ● The plot can become overcrowded if many ICE curves are drawn. ● It might not be easy to see the average.
LIME	● Able to work with diverse types of data. ● Very simple and easy to use. ● Provides local fidelity. ● Appropriate for the high quantity of explanatory variables. ● Human-friendly explanation.	● Explanations may vary significantly within small neighborhoods, which indicates instability. ● Possible to manipulate explanations to hide biases. ● Not capable of interpreting models characterized by non-linear decision boundaries.
Anchors	● Anchors are subsettable. ● Highly effective as it can be parallelized. ● Less computation compared to SHAP. ● Better generalizability compared to LIME.	● A highly configurable and impactful setup is required. ● Need discretization in several cases. ● Requires several calls to the ML model for building anchors. ● The concept of coverage is undefined in some areas.
SHapley Additive exPlanations (SHAP)	● Solid theoretical foundation in game theory. ● Fairly distributed prediction. ● It connects LIME and SHapley values. ● Fast implementation for tree-based models.	● KernelSHAP ignores feature dependence and is slow. ● TreeSHAP can produce unintuitive feature attributions. ● Misinterpretation and requires high computing time. ● Possible to generate misleading interpretations intentionally resulting hide biases.
Pixel Attribution (Saliency Maps)	● Visual and fast explanation to recognize images. ● There are many techniques available for pixel attribution to choose from.	● Very fragile. ● Highly unreliable. ● Shows Insensitivity to model and data.
Layer-wise Relevance Propagation (LRP)	● Scalable and provides explainability to complicated Deep Neural Networks (DNNs). ● Improve interpretability by calculating each neuron's weight.	● Performance can be sensitive to the specific architecture of the neural network. ● Show Vulnerability to Adversarial Attacks.
Grad-CAM	● Applies to many types of CNN models. ● Robust to adversarial perturbations. ● Provides model generalization by removing biases.	● Shows an inability to highlight intricate or detailed aspects. ● Aggregate individual interpretations of global knowledge are challenging.
Integrated Gradients	● Highly compatible with neural networks. ● Optimizes the heatmap to accurate and trustworthy explanations.	● Integrated gradients violate some of Shapley's axioms. ● Violation of symmetry and null effects.

Table 6
Different factors/considerations associated with different XAI techniques.

Name of XAI Techniques	Used	Data Type	Explanation Type
PDP	Commonly used in ML model analysis.	Tabular data	Global
ALE	Commonly used in ML model analysis, especially for describing the effect of individual features.	Tabular data	Global
Permutation Feature Importance	Often applied in feature selection and model interpretation.	Tabular data	Global
ICE	To understand the effect of a single feature on model predictions across different instances.	Tabular data	Local
LIME	To explain individual predictions of black-box ML models.	Tabular data, Text data, Image data	Local
Anchors	For providing interpretable conditions under which a model's prediction is highly likely to be correct.	Tabular data, Text data, Image data	Local
SHAP	To explain individual predictions of ML models.	Tabular data, Text data, Image data	Both local and global
Pixel Attribution (Saliency Maps)	To understand which pixels contribute most to the model's decision (Primarily in image classification tasks).	Image data	Local
LRP	In DL models to attribute relevance to input features.	Image data, Video data, Text data	Local
Grad-CAM	In CNNs, to understand which regions of an image contribute most to the model's decision.	Image data	Local
Integrated Gradients	To explain individual predictions of DL models	Image data, Text data	Local

search keywords, which may not cover all relevant terms, there is a high possibility of overlooking the potential published papers that were relevant and significant for the analysis. The second limitation is to restrict the search to only two databases (Scopus and IEEE Xplore Digital Library) for finding relevant research articles. Third, since studies only from the Q1 journal have been considered, there is a substantial possibility of omitting some of the valuable research works related to human disease diagnosis published in the other quartile of the journal.

8. Conclusions and future work

Considering the growing interest in XAI in recent years, an in-depth overview and investigation are performed through this review paper. This paper offered a systematic review of the 90 Q1 journals published between 2018 and September 27, 2023, that employ XAI techniques in various diseases diagnoses such as heart disease, cancer, diabetes, COVID-19, stroke, Alzheimer's disease, brain tumors, depression, stress, etc. Here, different XAI methods and their applications in disease diagnosis have been reported and analyzed. Various related topics of XAI have been covered throughout the paper, such as the XAI principles, the need for XAI, explanation properties with the elements of explanation styles, XAI and its stakeholder's roles, the accuracy vs. interpretability tradeoff, and the various challenges of considering XAI techniques in medical settings. This review article will help the researchers to understand the recent advancements in this field. Listing out the potential limitations of different XAI techniques will inspire researchers in this field to build new XAI approaches for medical settings where the consequences of actions are particularly significant and impactful. Since the use of AI in healthcare has been increasing rapidly, in the future, the ethical and societal implications of using XAI for human disease diagnosis can be addressed. Exploring issues such as the privacy of patients, algorithmic bias, and the potential impact on healthcare disparities can be crucial areas to investigate.

CRediT authorship contribution statement

Al Amin Biswas: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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